

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Use the classical electron energy-momentum relation for relativistic de Broglie wavelength :
 $\hat{p} = h / (\hat{\lambda} (2 m E)^{1/2} c),$
where h is Planck's constant (in erg Å·s), m is the electron mass (9.11×10^{-28} g), c is the speed of light (3.00×10^{10} cm/s), and $E = 1 \text{ eV} = 1.60 \times 10^{-12}$ erg .

Calculate the factor :
 $(2 m E / c^2)^{1/2} = [(2 \times 9.11 \times 10^{-28} \times 1.60 \times 10^{-12}) / (3.00 \times 10^{10})^2]^{1/2} = 1.875 \times 10^{-23} \text{ g Å·cm/s}.$

Then $\hat{\lambda} = h / (\hat{p} (2 m E)^{1/2} c) = (6.625 \times 10^{-27} \text{ erg Å·s}) / (1.875 \times 10^{-23} \text{ g Å·cm/s} \times 3.00 \times 10^{10} \text{ cm/s}) = 1.23 \times 10^{-7} \text{ cm}.$

Correct answer